



University of Al-Ameed
College of Medicine
Department of Anatomy



HEAD AND NECK

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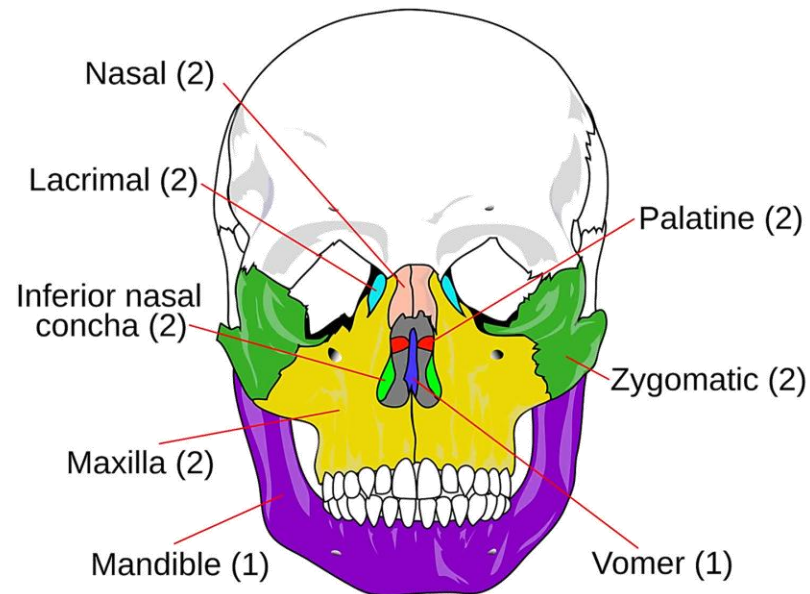
M.B.Ch.B. M.Sc. Ph.D. Human Anatomy

Objectives

1. Identify the detailed anatomy of the skull.
2. Scalp and Face
3. Temporal region and infratemporal fossa
4. Illustrate the definitive anatomy of the neck
5. Root of the neck
6. Nose and paranasal sinuses
7. Orbit and eye
8. Oral cavity and salivary glands
9. Pharynx and larynx
10. Ear anatomy

SKULL

- Head skeleton.
- 22 bones.
- Cranium + Mandible
- Cranium = Calvaria + Face
- Calvaria = 4 unpaired + 2 paired
- Face = 6 paired + 2 unpaired

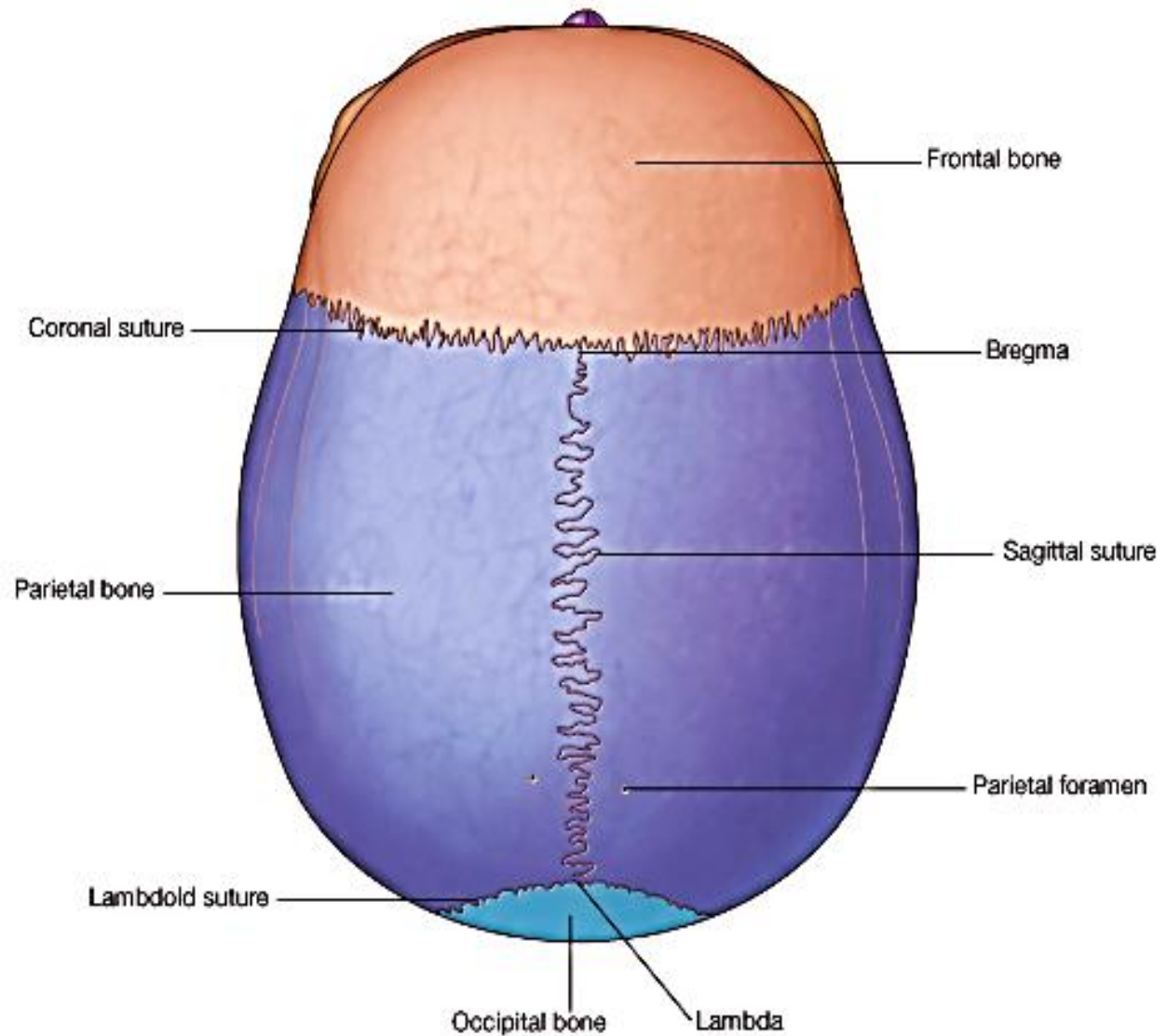


14 Facial Bones



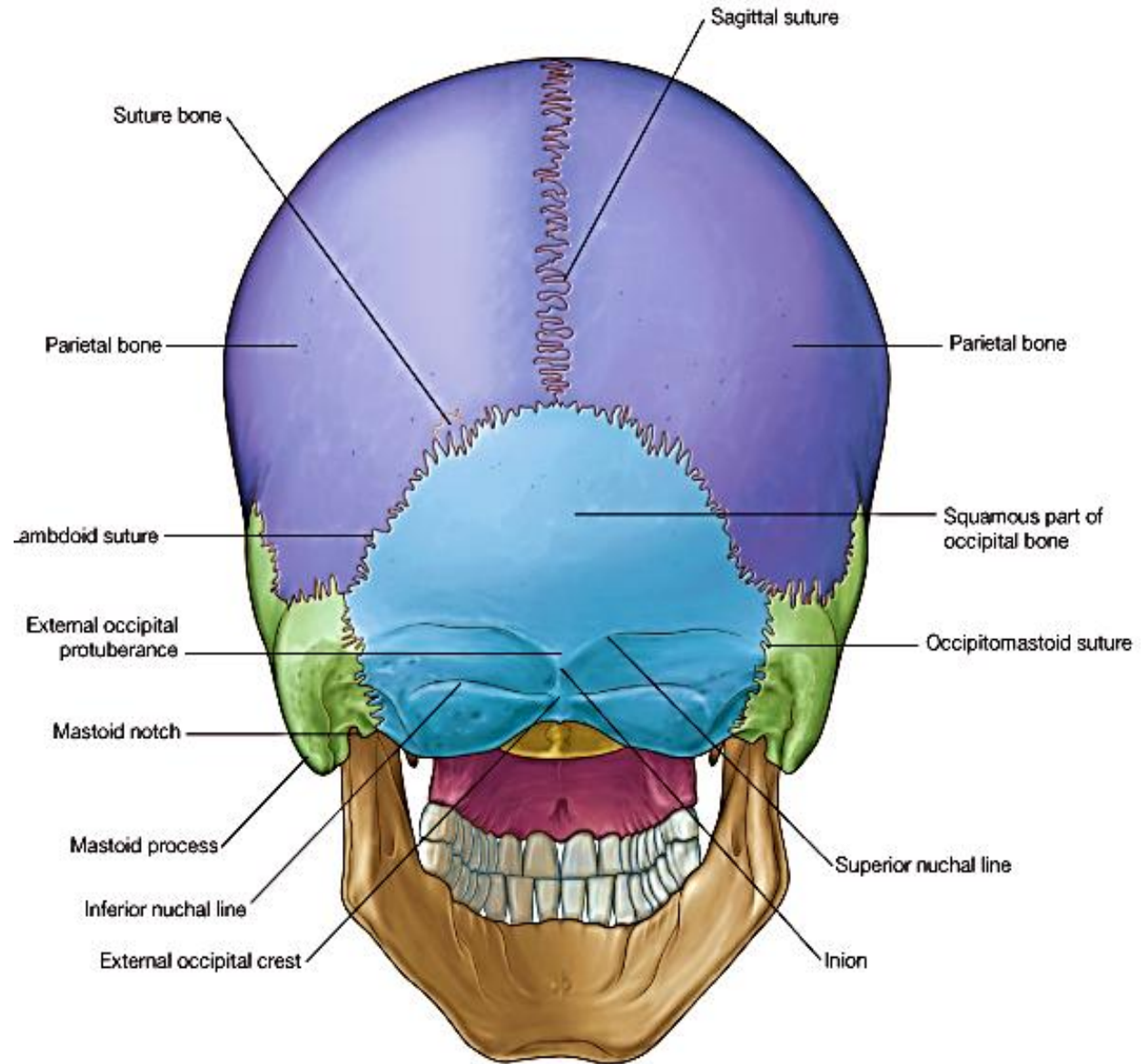
Superior View (Norma Verticalis)

- Skull cap (vault)
- Sutures;
 - A. Sagittal
 - B. Coronal
 - C. Lambdoid
- Bregma
- Lambda



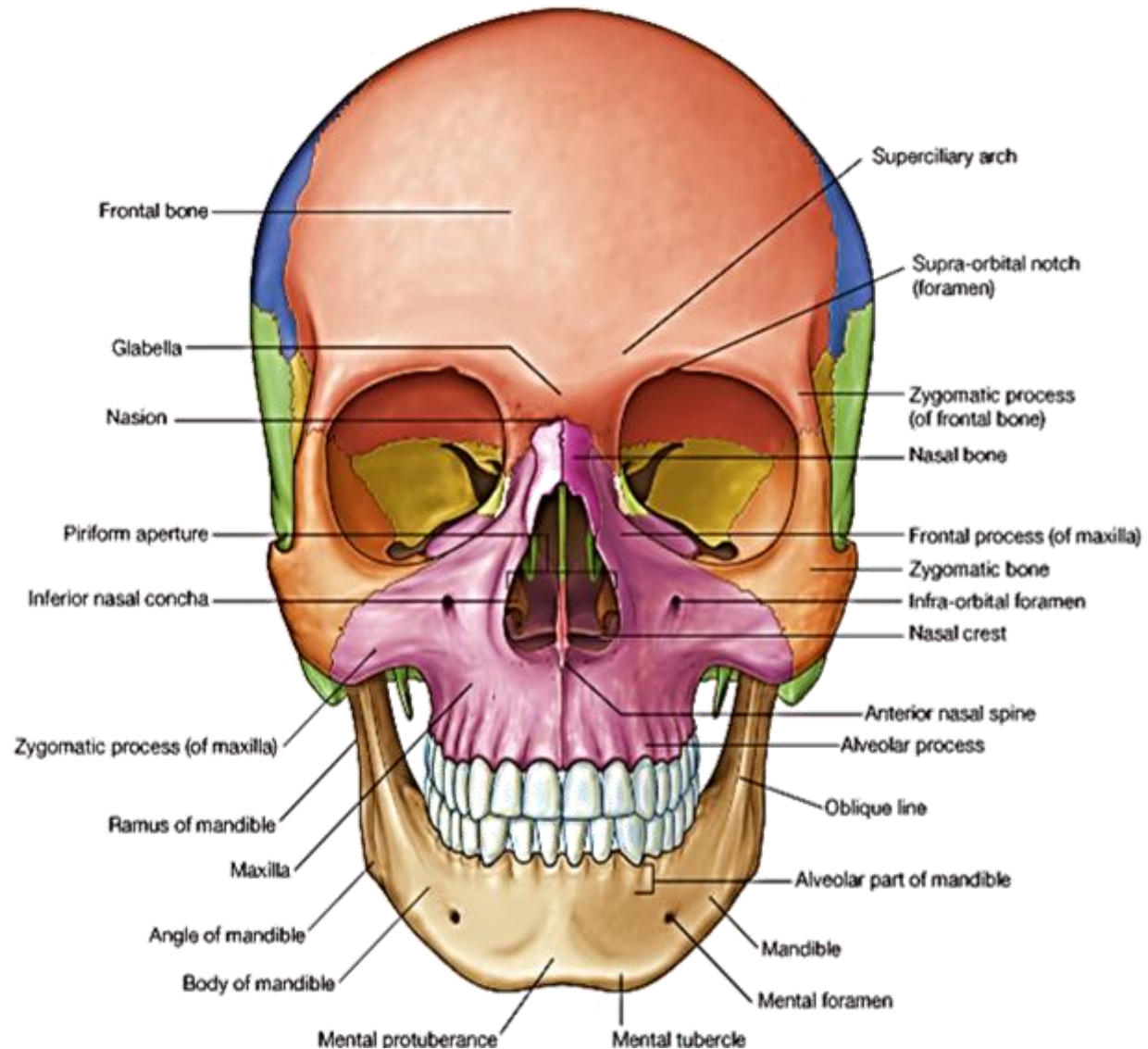
Posterior View (Norma Occipitalis)

- Sutures: sagittal, lambdoid, occipitomastoid
- External occipital protuberance
- Inion
- Nuchal lines
- Mastoid notch



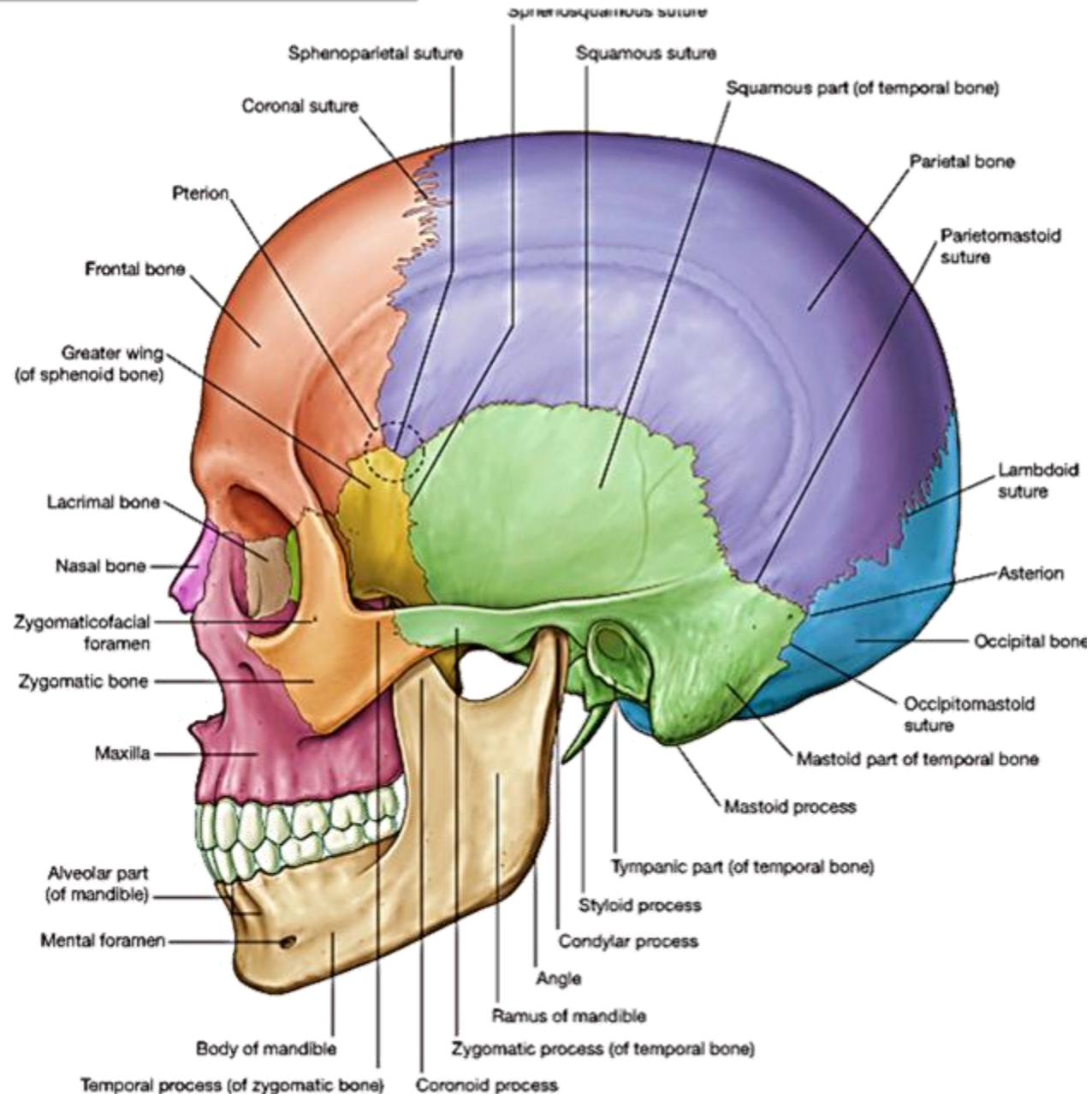
Anterior View (Norma Frontalis)

- Forehead
- Superciliary arch
- Glabella
- Nasion
- Orbit
- Supraorbital foramen
- Infraorbital foramen
- Mental foramen
- Piriform aperture
- Maxilla
- Mandible



Lateral View (Norma Lateralis)

- Sutures
 - Coronal
 - Lambdoid
 - Sphenoparietal
 - Sphenotemporal
- Pterion
- Temporal lines
- Zygomatic arch
- Temporal bone



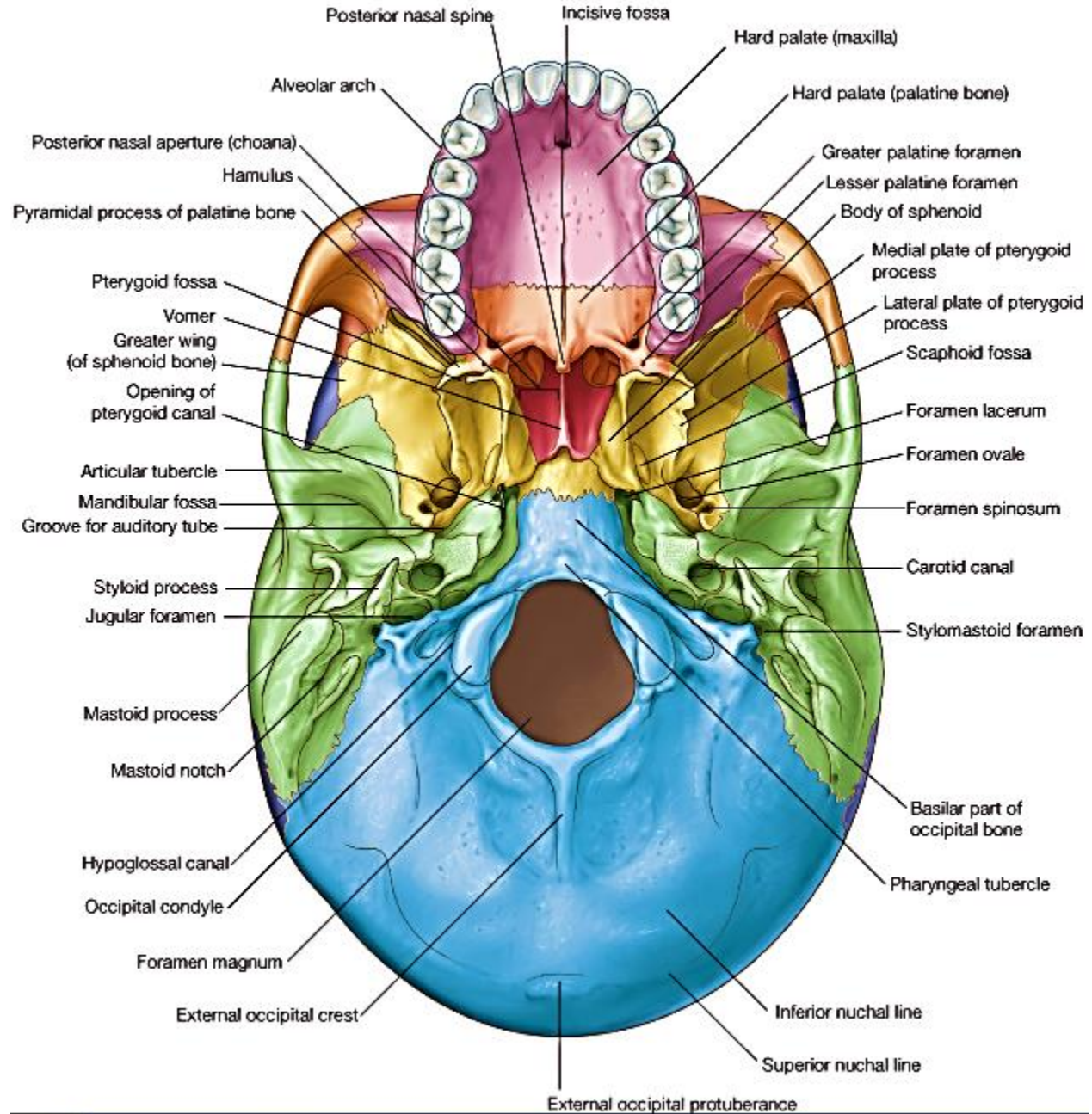
Inferior View (Norma Basalis)

- From middle incisor to superior nuchal line, and to mastoid processes and zygomatic arches.
- For descriptive purposes it can be divided into three parts:

A. Anterior

B. Middle

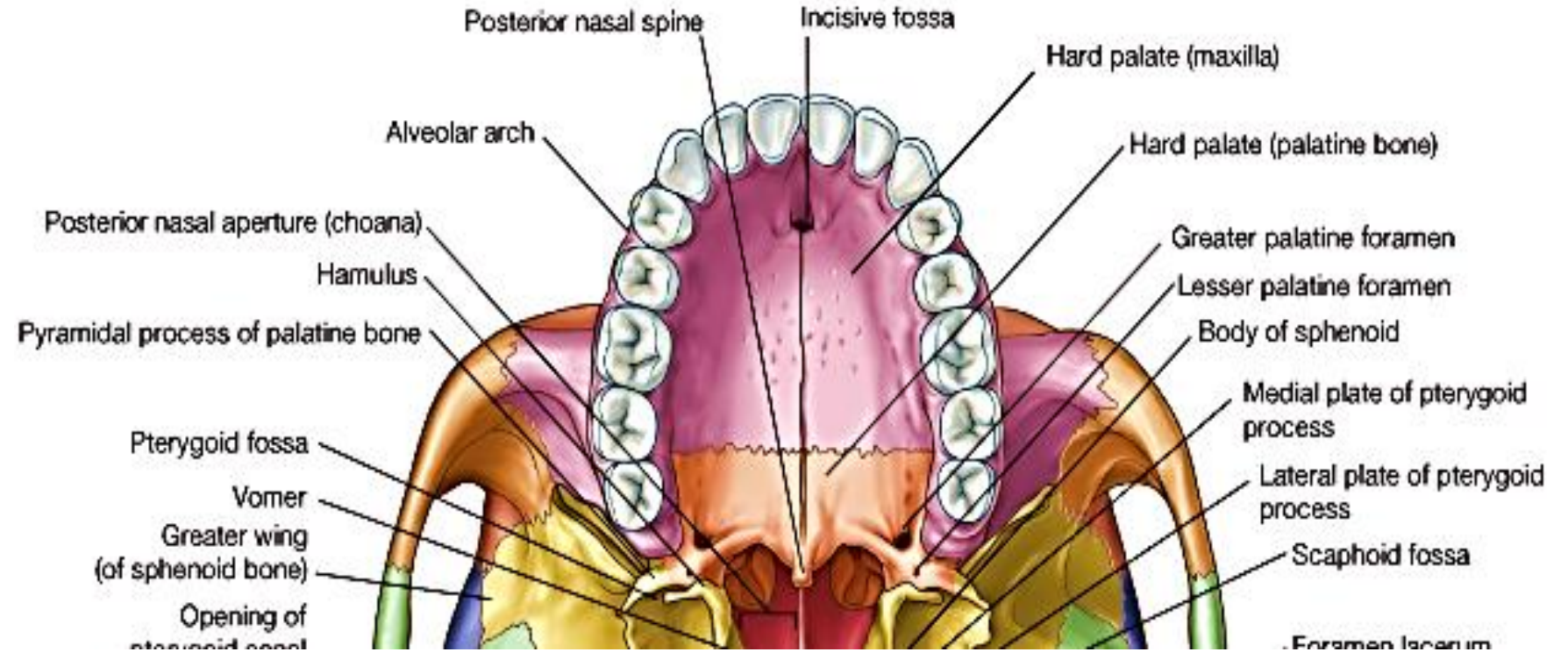
C. Posterior



Inferior View (Norma Basalis)

A. Anterior part:

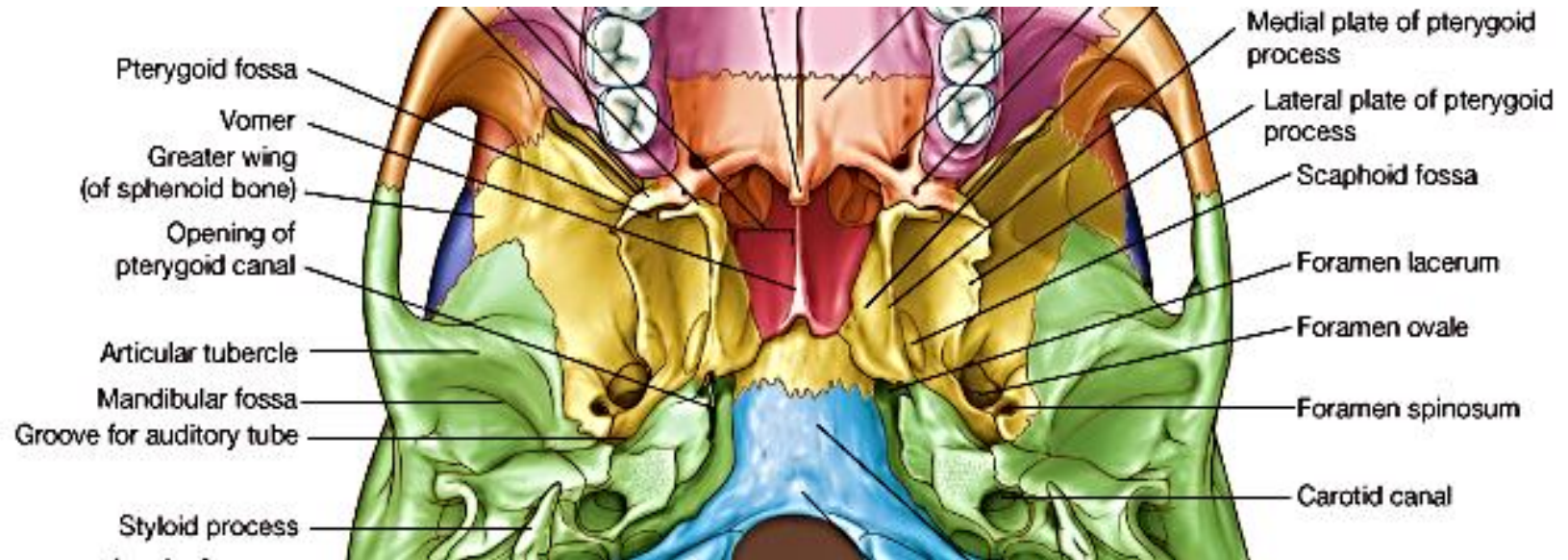
- Teeth
- Hard palate
- Incisive fossa
- Palatine foramina
- Posterior nasal spine



Inferior View (Norma Basalis)

B. Middle Part: behind hard palate to foramen magnum

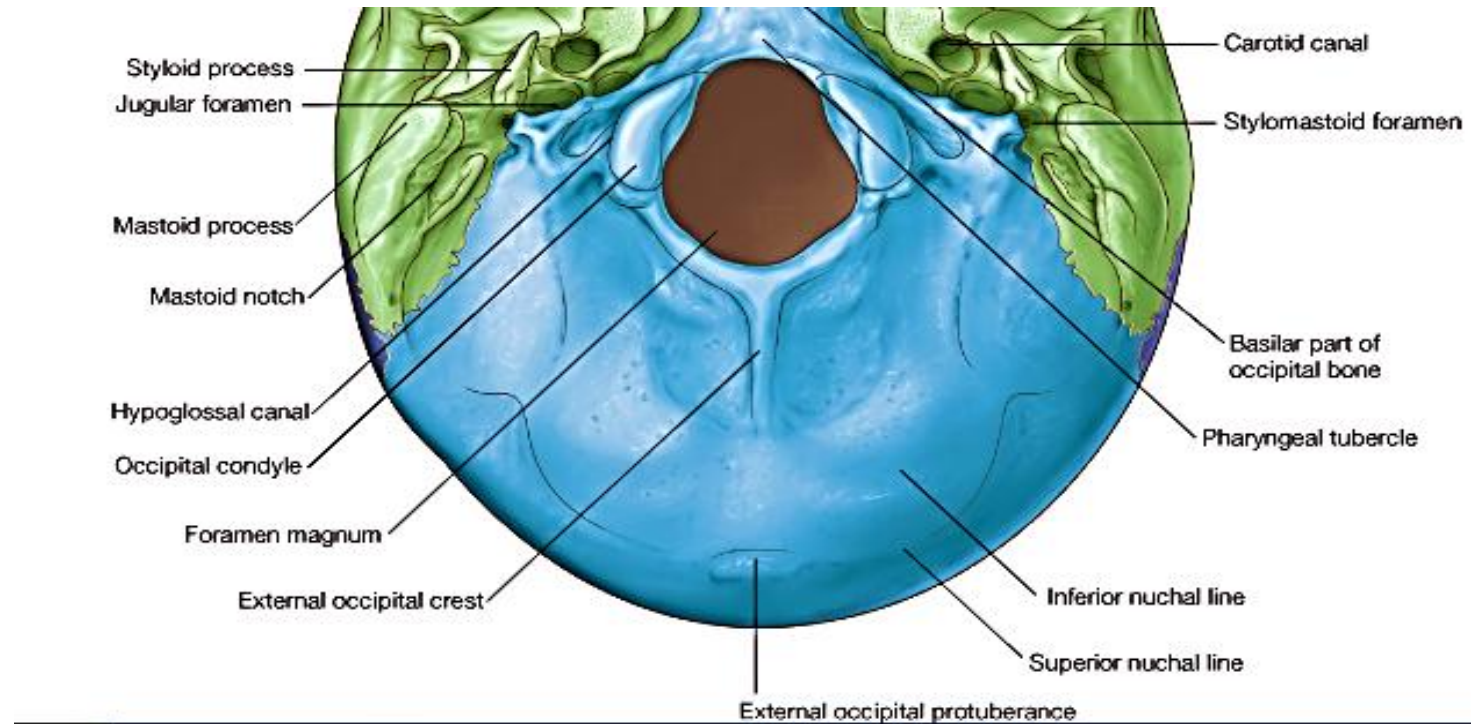
- Vomer & Sphenoid bones
- Sphenoid (Body, greater and lesser wings, pterygoid processes, and spine)
- Foramina (spinosum, ovale, lacerum, and carotid canal)
- Basilar part of occipital bone
- Pharyngeal tubercle
- Mandibular fossa



Inferior View (Norma Basalis)

C. Posterior Part: from the anterior edge of the foramen magnum to the superior nuchal lines.

- Basilar part of occipital bone (foramen magnum, condyles)
- Temporal bone (mastoid and styloid processes)
- Jugular foramen
- Hypoglossal canal
- Stylomastoid foramen



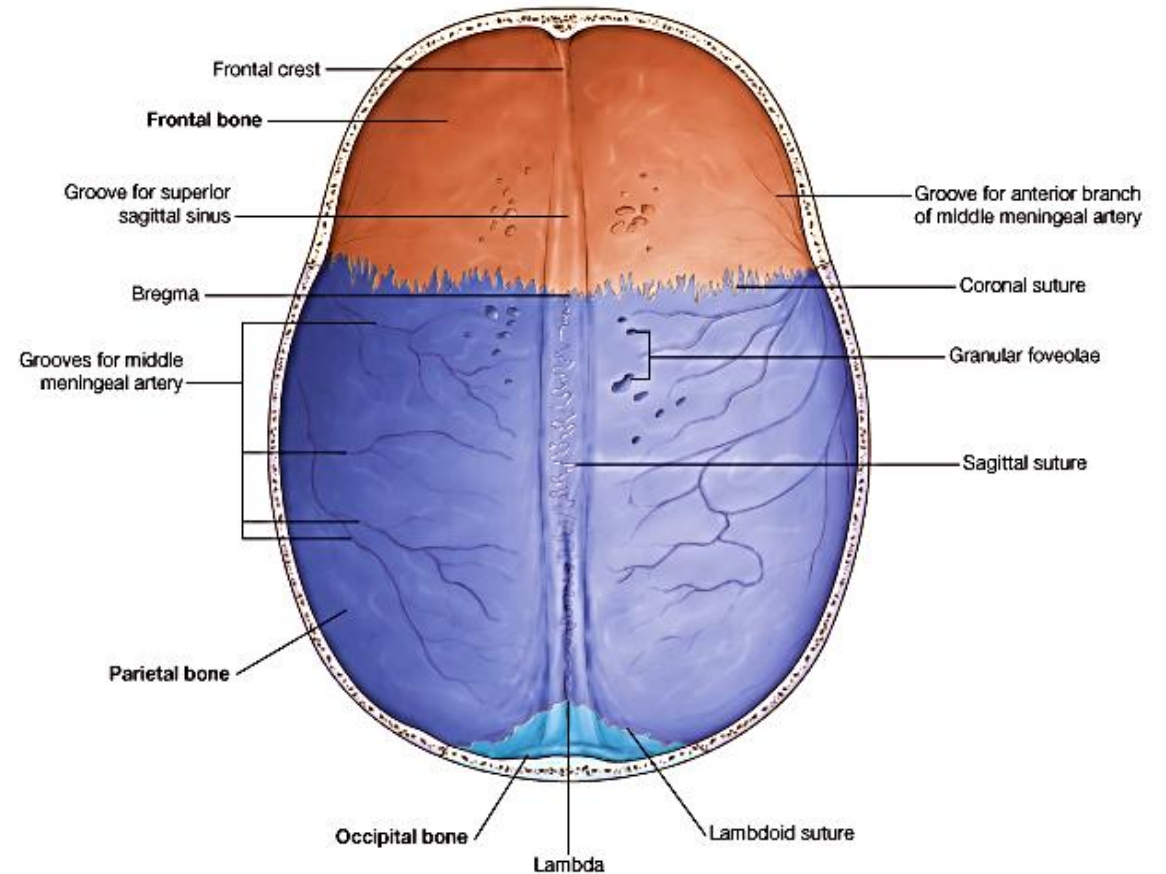
External foramina of the skull

Foramen	Structures passing through foramen
ANTERIOR VIEW	
Supra-orbital foramen	Supra-orbital nerve and vessels
Infra-orbital foramen	Infra-orbital nerve and vessels
Mental foramen	Mental nerve and vessels
LATERAL VIEW	
Zygomaticofacial foramen	Zygomaticofacial nerve
SUPERIOR VIEW	
Parietal foramen	Emissary veins

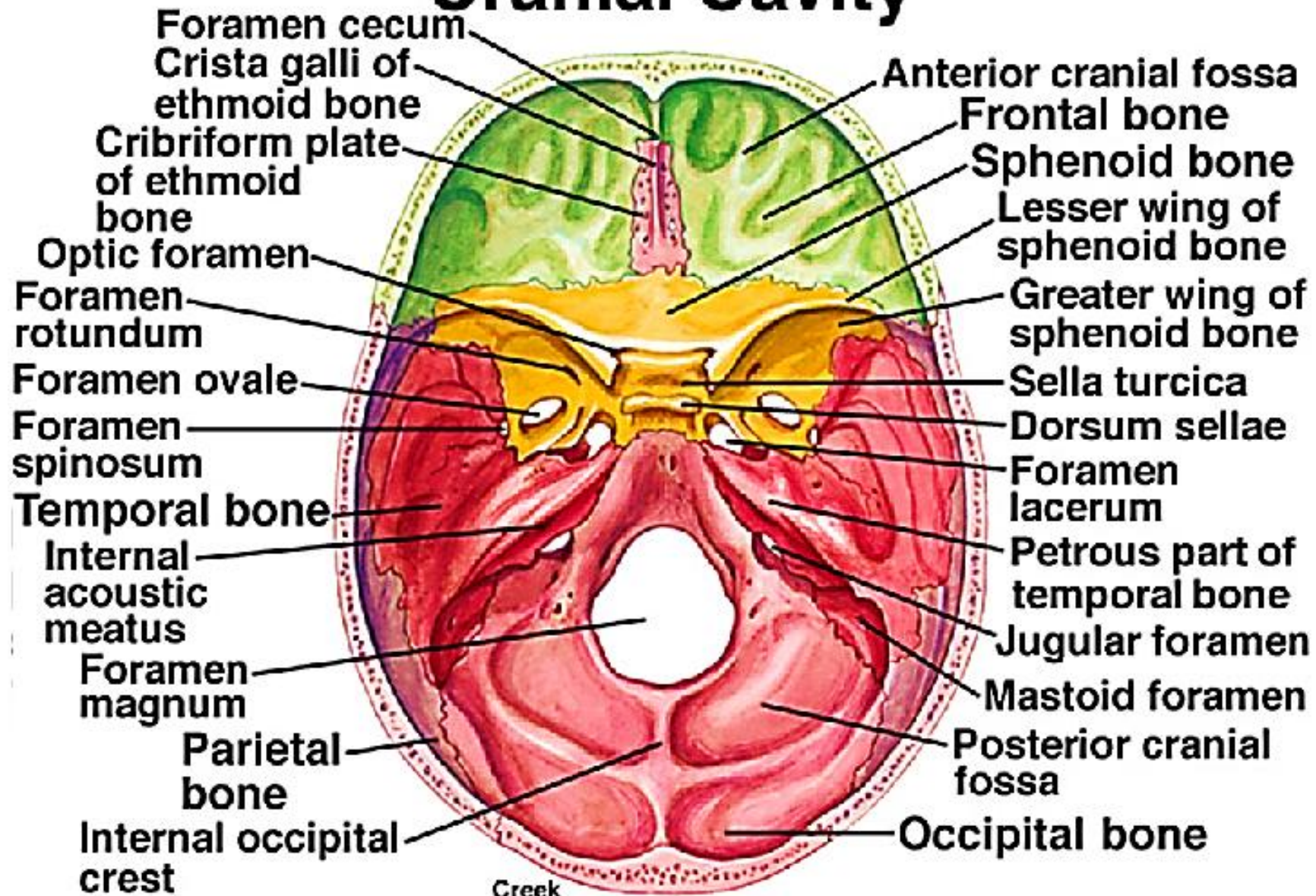
Foramen	Structures passing through foramen
INFERIOR VIEW	
Incisive foramina	Nasopalatine nerve; sphenopalatine vessels
Greater palatine foramen	Greater palatine nerve and vessels
Lesser palatine foramina	Lesser palatine nerves and vessels
Pterygoid canal	Pterygoid nerve and vessels
Foramen ovale	Mandibular nerve [V ₃]
Foramen spinosum	Middle meningeal artery
Foramen lacerum	Filled with cartilage
Carotid canal	Internal carotid artery and nerve plexus
Foramen magnum	spinal cord; vertebral arteries and nerve plexuses; anterior spinal artery; posterior spinal arteries; roots of accessory nerve
Condylar canal	Emissary veins
Hypoglossal canal	Hypoglossal nerve [XII] and vessels
Jugular foramen	Internal jugular vein; inferior petrosal sinus, IX, X, XI CN.
Stylomastoid foramen	Facial nerve [VII]

CRANIAL CAVITY

- It is the space within the calvaria that contains the brain, meninges, proximal parts of the cranial nerves, blood vessels, and cranial venous sinuses.
- **Roof :**
- Sutures
- Frontal crest
- Vascular grooves

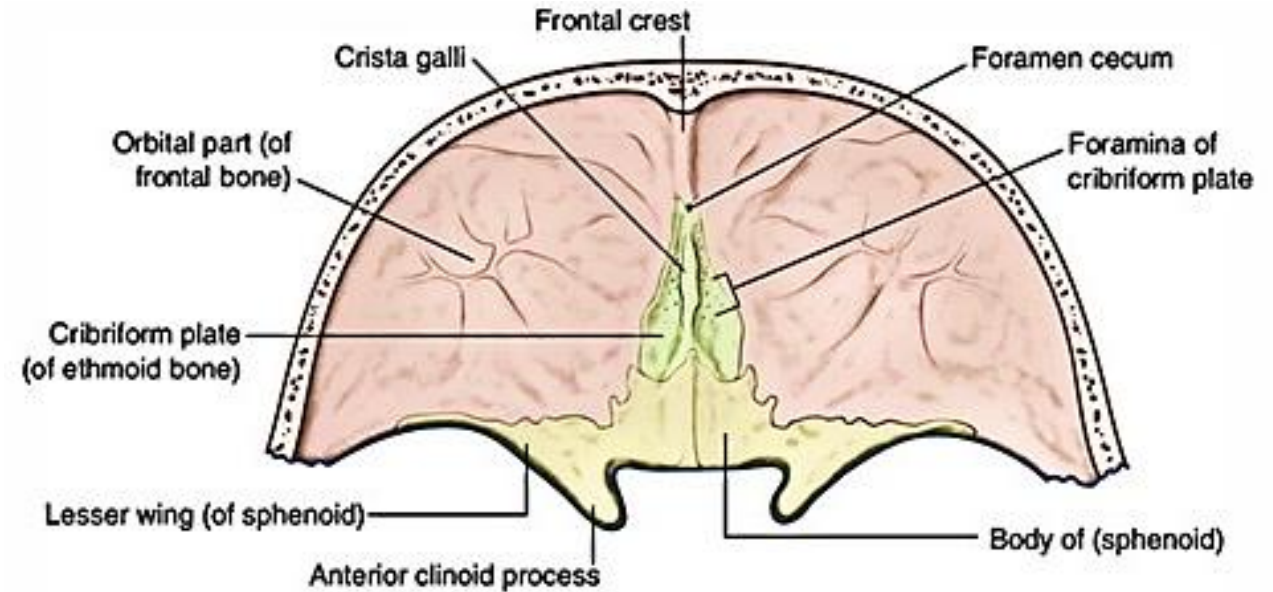


Cranial Cavity



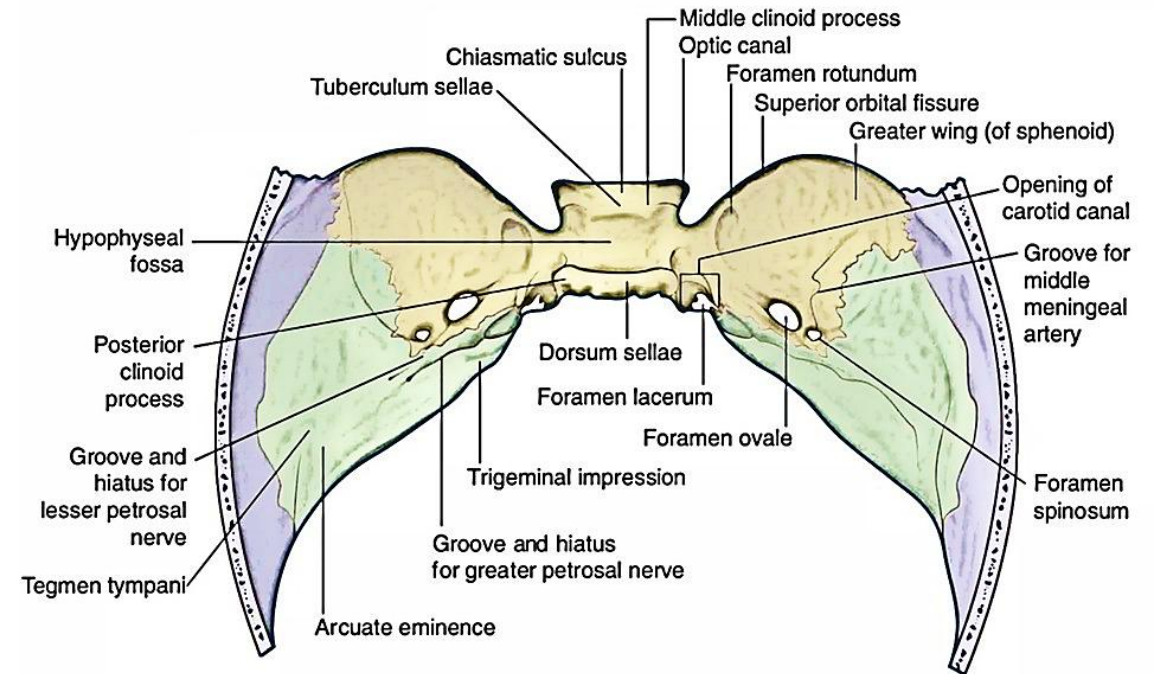
Anterior Cranial Fossa

- It is above the nasal cavity and the orbits, and it is filled by the frontal lobes of the cerebral hemispheres.
- Frontal crest
- Foramen cecum
- Crista galli
- Cribriform plate of the ethmoid
- Anterior clinoid processes



Middle Cranial Fossa

- The boundary between the anterior and middle cranial fossae in the midline is the anterior edge of the chiasmatic sulcus, which is a smooth groove stretching between the optic canals across the body of sphenoid.
- The posterior boundaries of the middle cranial fossa are formed by the anterior surface of the petrous and mastoid part of temporal bone.
- Sella turcica (hypophysial fossa)
- Middle and posterior clinoid processes
- Superior orbital fissure
- Foramen rotundum, ovale, spinosum, carotid canal and foramen lacerum
- Tegmen tympani



Posterior Cranial Fossa

- It is the largest and deepest of the three cranial fossae and contains the brainstem (midbrain, pons, and medulla) and the cerebellum.
- The anterior boundaries of the posterior cranial fossa in the midline are the dorsum sellae and the clivus. Posteriorly the squamous part of the occipital bone.
- Foramen magnum
- Internal acoustic meatus
- Jugular foramen and tubercle
- Hypoglossal canal
- Condylar canal

